SNOWFLAKE ADVANCED DATA ENGINEERING

Lab 01 | Prepare the event source

Start with reproducible input; no processing pipeline exists yet.

Existing Added / changed Control Evidence

The source reports full order states as immutable events. The source includes an identical e2 replay, an invalid e3 amount, and a late older revision.

EXPECTED CHECKPOINT

The staged files are ready for ingestion in Lab 02.

This is SQL-based source preparation. No instructor-only Python setup is required.

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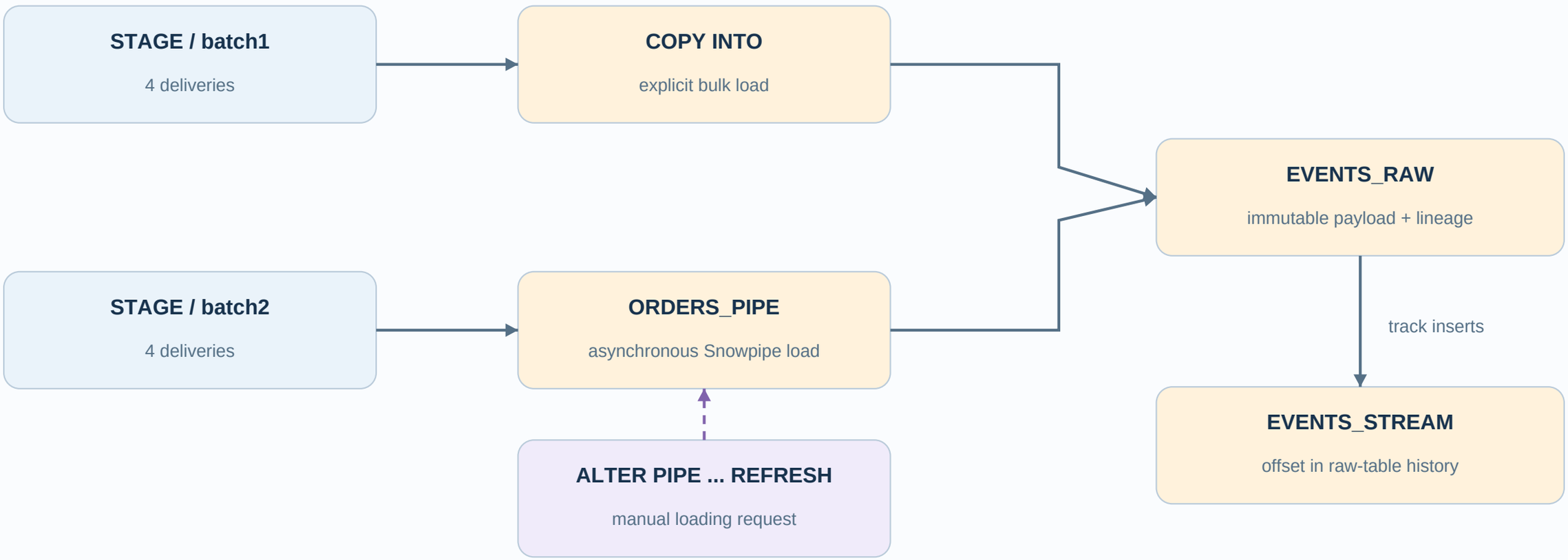
Lab 02 | Load Bronze and track new rows

Add two ingestion routes and one append-only stream.

Existing Added / changed Control Evidence



EXPECTED CHECKPOINT

Raw: 8 deliveries | Pending stream: 8 rows

AUTO_INGEST = FALSE in this lab. REFRESH requests loading; it does not wait for the file to finish loading.

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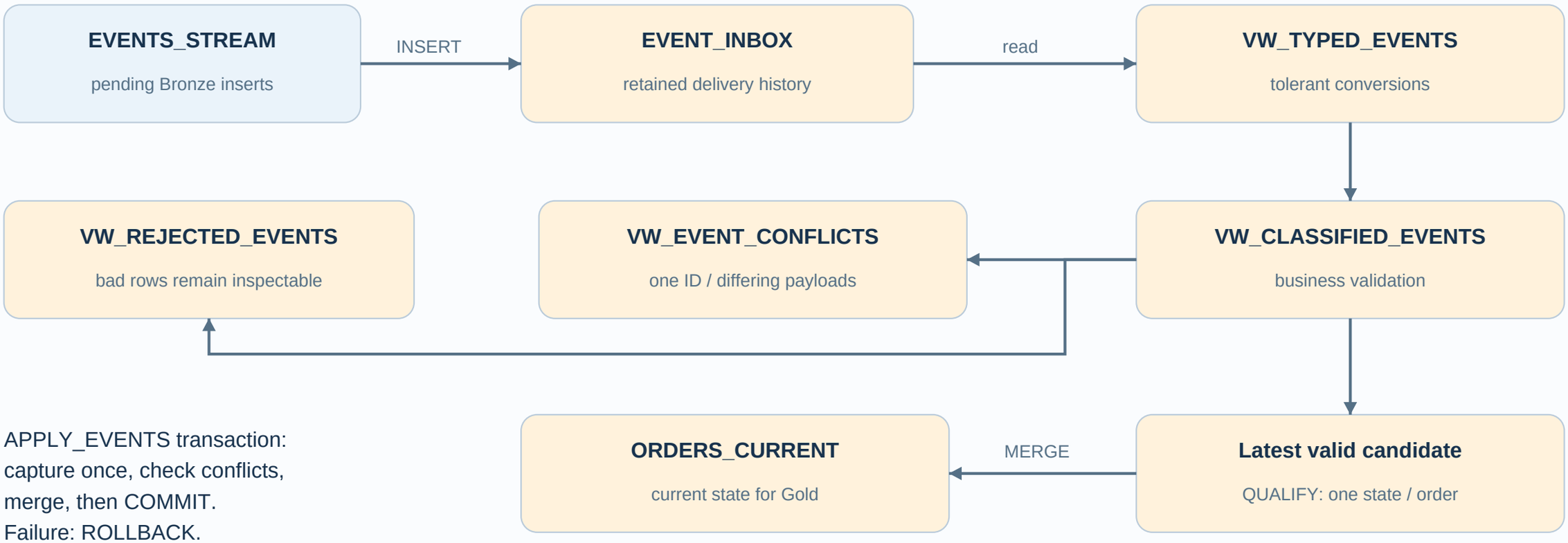
Lab 03 | Develop the Silver transformation

Define the destinations first; build and inspect the processing steps.

Existing Added / changed Control Evidence



APPLY_EVENTS transaction:
capture once, check conflicts,
merge, then COMMIT.
Failure: ROLLBACK.

EXPECTED CHECKPOINT

Inbox: 8 | Current orders: 3 | Rejects: 1 | Pending: 0 | Active value: USD 150

Views do not store copies. Identical replays remain in the inbox; only the latest accepted state is merged.

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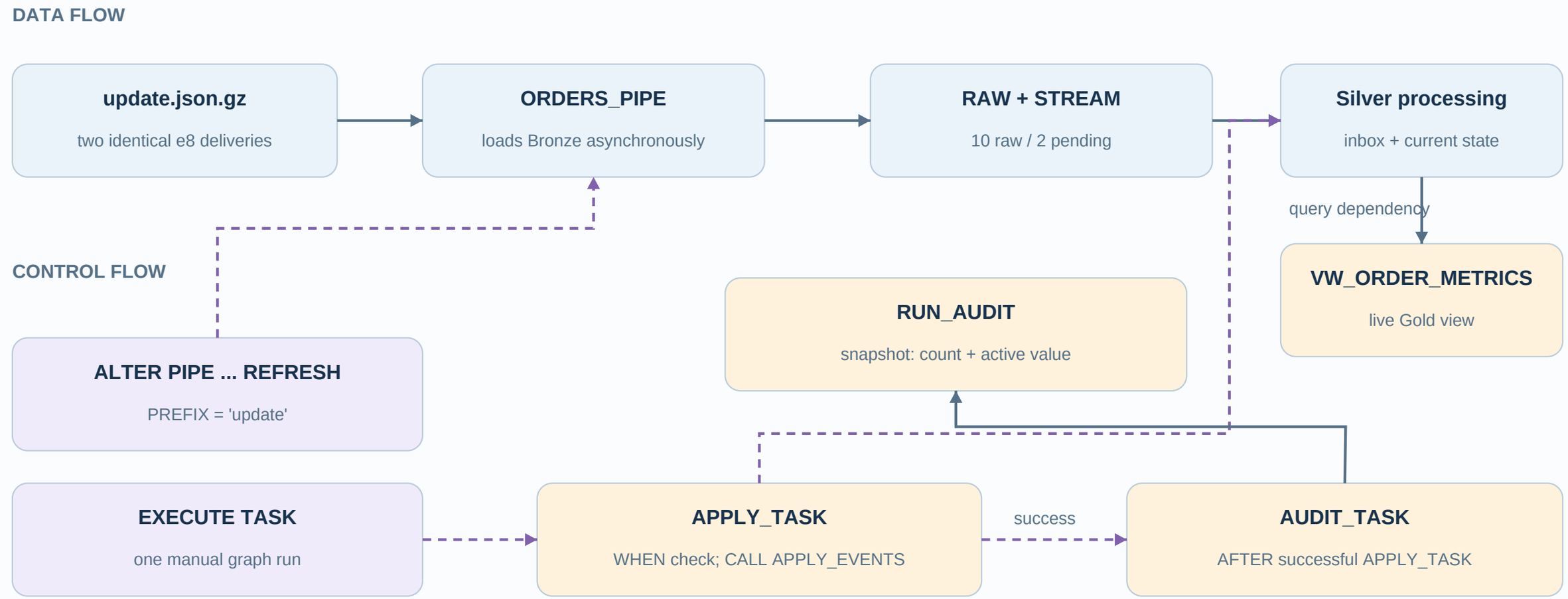
Lab 04 | Orchestrate processing and expose Gold

Two separate control requests: load the file, then execute the task graph.

Existing Added / changed Control Evidence



EXPECTED CHECKPOINT

Raw / inbox: 10 | Current: 3 | Pending: 0 | US active orders: 2 / USD 165

Root stays suspended; child is resumed. Wait for the corresponding graph run to finish. Gold reads current state; audit stores a snapshot.

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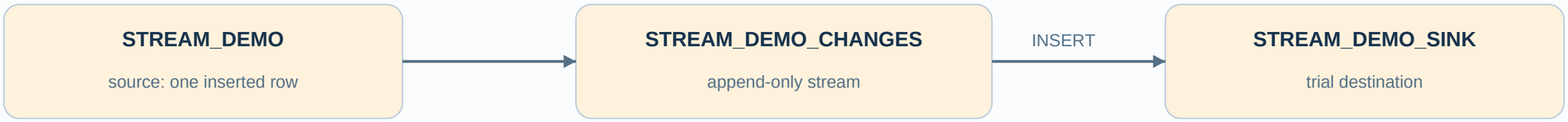
Lab 05 | Prove correctness and rollback

Add evidence; keep the demonstration separate from the order pipeline.

Existing Added / changed Control Evidence

SEPARATE DEMONSTRATION OBJECTS



BEGIN TRANSACTION -> INSERT from stream -> inspect -> ROLLBACK

EXPECTED CHECKPOINT

After rollback: demonstration stream = 1 row | Demonstration sink = 0 rows

Run acceptance before Lab 06. A rollback leaves the stream offset unchanged; it does not delete the original source row.

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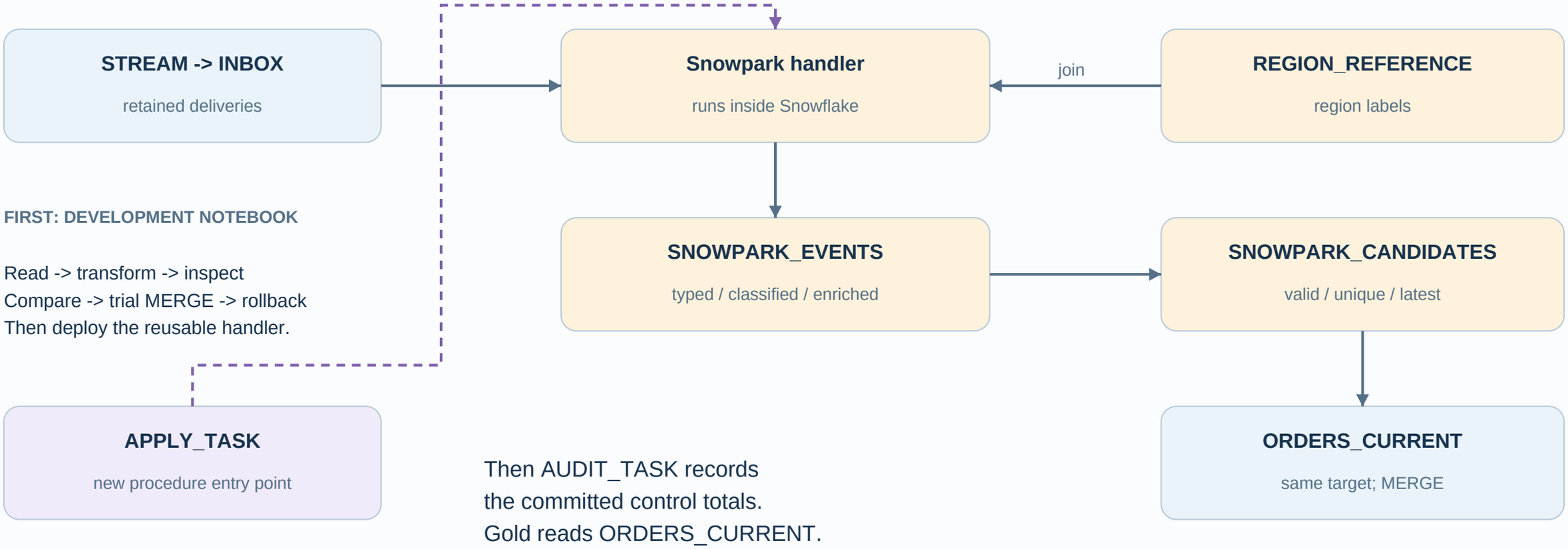
Lab 06 | Develop and deploy with Snowpark

Notebook: inspect each transformation, trial writes with rollback, then assemble the handler.

Existing Added / changed Control Evidence



EXPECTED CHECKPOINT

Raw / inbox: 11 | Current: 3 | Rejects: 1 | Pending: 0 | Active value: USD 165

The synthetic replay is inserted directly into Bronze to isolate business idempotency. It does not retest file ingestion.

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